

Suggestion System Architecture (Design Definition)

Health Data Bank System

1. Purpose

This document defines the architecture of the Suggestion System within the Health Data Bank. The purpose of this system is to generate meaningful, data-driven suggestions for users based on their health data, activity patterns, and system-generated insights.

This document establishes the structure, flow, and behavior of the system while avoiding implementation-specific details.

2. Context

The Suggestion System is part of the system's advanced features for automated insights. It enhances user engagement by analyzing user data and providing relevant recommendations.

This system integrates with existing components such as health data entries, reporting services, dashboards, and audit logging.

3. Design Objectives

The Suggestion System should:

- Generate suggestions based on user data trends and patterns
- Support rule-based and data-driven analysis
- Integrate with reporting and aggregated data
- Deliver suggestions through the user dashboard
- Maintain role-based access control and data privacy
- Support future extensions for more advanced analytics

4. Trigger Conditions

The suggestion system should be triggered under the following conditions:

- Periodic analysis of user data at defined intervals
- Submission of new health data or form entries

These triggers ensure that suggestions remain timely and relevant.

5. Preconditions

Before generating suggestions, the system must ensure:

- The user has sufficient historical data for meaningful analysis
- The suggestion feature is enabled in the user's settings

If these conditions are not met, suggestion generation should not proceed.

6. System Components

6.1 Data Sources

The system uses the following data sources:

- User health data entries
- Aggregated and analyzed metrics from reporting services
- Historical trends and activity records

These sources provide the basis for generating suggestions.

6.2 Suggestion Rules

Suggestion rules define when a suggestion should be generated. These rules may include:

- Threshold-based conditions (e.g., unusual values)
- Trend-based analysis (e.g., increasing or decreasing patterns)
- Missing or incomplete data

The system should allow flexible rule definitions for future expansion.

6.3 Suggestion Generation Process

The system evaluates available data against suggestion rules to:

- Identify relevant patterns or conditions
- Determine whether a suggestion should be generated
- Create suggestions when conditions are satisfied

This defines expected behavior without specifying implementation.

6.4 Suggestion Structure

Each suggestion should include:

- User identifier
- Suggestion message
- Suggestion type (e.g., recommendation, alert)
- Optional reference to related data
- Timestamp

This ensures consistent handling across the system.

6.5 Suggestion Storage

Suggestions should be stored in a structured format to support:

- Retrieval for dashboard display
- Tracking of suggestion history
- Future analysis and validation

6.6 Suggestion Delivery

Suggestions should be delivered primarily through:

- User dashboard display

Optional integration with the notification system may be used for high-priority suggestions.

6.7 Integration with Reporting System

The suggestion system should integrate with reporting and aggregation services to:

- Use computed metrics and trends
- Ensure accuracy and consistency
- Align suggestions with system-generated reports

7. System Flow

The system behavior follows this sequence:

- The system is triggered by periodic analysis or new data submission
- Preconditions are validated
- User data and aggregated metrics are analyzed
- Suggestion rules are evaluated
- Users meeting suggestion conditions are identified
- Suggestions are generated
- Suggestions are stored
- Suggestions are displayed on the user dashboard

8. Example Scenario

If a user's data shows a declining trend:

- The system detects the pattern through analysis
- A suggestion rule is satisfied
- A suggestion is generated (e.g., "Your recent data shows a decline. Consider reviewing your health inputs.")
- The suggestion is stored and displayed on the dashboard

9. Alternate Flows

The system should handle the following cases:

- **Insufficient data:** Suggestion generation is skipped and the reason is recorded
- **Feature disabled:** No suggestions are generated for the user
- **Analysis error:** The error is recorded and the process may be retried later

10. Role-Based Access Control (RBAC)

The system must ensure:

- Suggestions are visible only to the relevant user
- Access is restricted based on user roles
- Sensitive information is not exposed to unauthorized roles
- Authorized roles (e.g., providers) may access limited summary views where applicable

11. Postconditions

After suggestion generation:

- Suggestions are available on the user dashboard
- Suggestion events are recorded for tracking and auditing

12. Design Constraints

The suggestion system must:

- Integrate with existing health data and reporting components
- Maintain privacy and security requirements
- Support scalability and future enhancements
- Remain independent of specific implementation details

13. Conclusion

This document defines the architecture of the suggestion system by outlining its triggers, components, data flow, and constraints. It provides a structured foundation for generating data-driven suggestions while ensuring consistency with the overall system design.

